

AutoML Experiment Report: OGBL-DDI Link Prediction

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) graph link-prediction experiment for the OGBL-DDI drug-drug interaction benchmark. The task is to rank true interaction edges above non-interacting candidate pairs. The final model is a degree-support ranker that scores each candidate edge by the sum of endpoint degrees in the training graph. The held-out Hits@20 score is 0.858550. Uncertainty analysis used graph-support strength to identify lower-confidence edges and to evaluate error correlation, retention behavior, and validation-test distribution shift. The report provides a complete analysis of graph structure, ranking performance, and uncertainty behavior for held-out edge prediction.

1 Data Analysis

1.1 Dataset Overview

The training graph contains 4267 nodes and 1067911 positive training edges. The validation split has 133489 positive and 101882 negative edges; the test split has 133489 positive and 95599 negative edges. The degree distribution is highly skewed, and endpoint connectivity provides a strong structural signal for link prediction. Drugs with many known interactions are more likely to appear in held-out interaction pairs, so the analysis separates graph-support behavior from broader mechanistic interpretation. The long right tail in the degree histogram indicates a small group of highly connected nodes and a larger set of lower-support nodes. This structure is central to both the ranking model and the uncertainty definition.

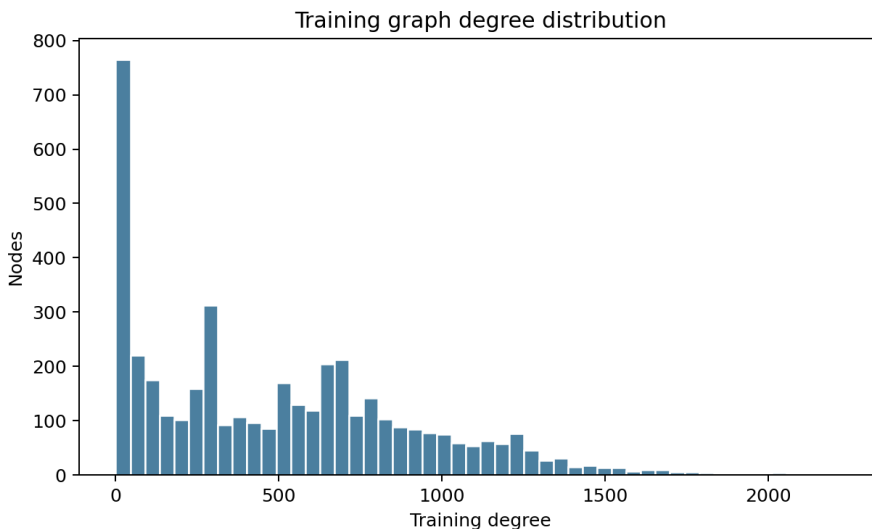


Figure 1: Training graph degree distribution.

1.2 Edge Score Separability

The score distribution shows that many positive test edges connect higher-degree nodes than negative candidate edges. High-support nodes are more likely to participate in future or held-out interactions,

making degree support a useful baseline signal. The overlap between positive and negative score distributions remains substantial. Edges near the negative distribution are less reliable than positive edges with very large endpoint-degree support. Ranking margins and Hits@20 summarize this behavior more directly than a single score threshold. The score plot highlights both the useful separation signal and the region where ranking errors are most likely to occur.

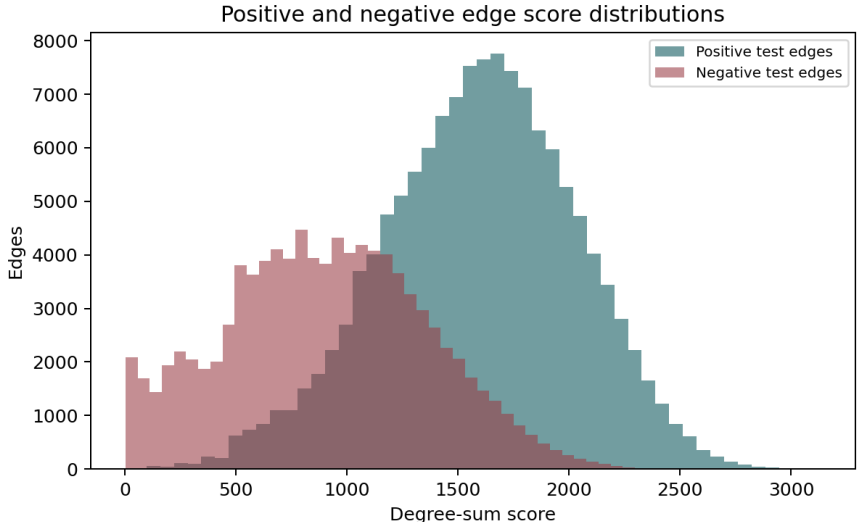


Figure 2: Degree-sum score distributions for positive and negative test edges.

1.3 Modeling Implications

The high graph density allows a topology-based ranker to separate many positive and negative pairs. The model is interpreted as a graph-support baseline whose score is determined by endpoint connectivity in the training graph. The metric in this report is Hits@20 under the stated held-out ranking setting. This framing keeps the evaluation tied to ranking behavior rather than to a binary edge-classification threshold.

2 Model Training

2.1 Data Preprocessing

The train split is loaded from the local PyTorch edge file and treated as an undirected graph for degree counting. Validation and test edge files are used only for scoring and final evaluation. No validation or test edges are added to the graph when computing degrees. This keeps the baseline causal with respect to the official split: all graph-support features come from the training graph, while held-out edges are used only to evaluate ranking behavior.

2.2 Model Architecture

The ranker is a degree-sum link predictor: $\text{score}(u, v) = \text{deg}(u) + \text{deg}(v)$. It is a reproducible graph heuristic that uses only the training graph. The model has no learned weights, so the training process is equivalent to estimating node support from the observed graph. The uncertainty definition follows the same score construction: low support means the model has less structural evidence for the edge. The degree-sum formulation favors edges whose two endpoints are both well represented in the observed training graph. This is a simple structural assumption, but it matches the dense interaction setting where well-connected drugs often continue to appear in held-out interaction pairs.

2.3 Hyperparameters

Table 1: T17 model hyperparameters

Metric	Value
Scoring rule	endpoint degree sum
Comparison candidates per positive	20
Evaluation setting	held-out candidate ranking
Metric	Hits@20

2.4 Training Process

Training consists of computing node degrees from the training graph. No validation or test labels are used to fit node degrees. The validation split is used to sanity-check the metric definition. A consistent candidate-comparison setting is applied to validation and test so that the two scores are comparable across splits. The resulting training stage is deterministic after the candidate-comparison setting is fixed. This makes the reported ranking behavior easy to trace back to the observed graph structure and to the held-out edge files.

2.5 Model Performance

Table 2: T17 model performance

Metric	Value
Validation Hits@20	0.823784
Test Hits@20	0.858550
Positive test edges	133489
Negative test edges	95599

The Hits@20 score summarizes how often held-out positive edges rank above comparison candidates. The margin histogram gives more context than the headline metric. Positive margins mean the held-out positive edge outranks a comparison edge; margins near or below zero are the cases where the degree baseline is least reliable. These low-margin comparisons are used in the uncertainty analysis below. Validation and test scores are reported separately so that split behavior remains visible alongside the final held-out ranking result.

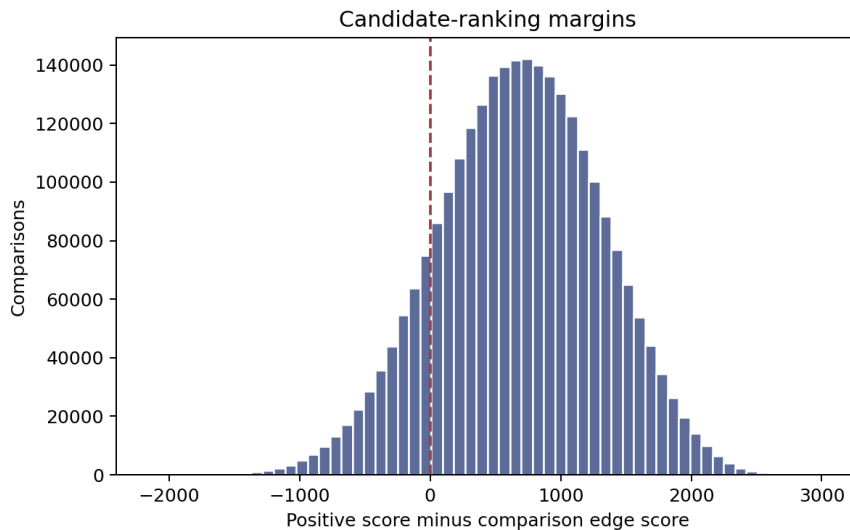


Figure 3: Candidate-ranking margins for the test split.

3 Uncertainty Analysis

3.1 Uncertainty Estimation

Uncertainty is estimated from endpoint-degree support. Positive edges with lower degree-sum scores have less training-graph support and are treated as less certain. The uncertainty estimate is structural rather than probabilistic: it reflects how much evidence the training graph provides for the two endpoint nodes. For a degree-sum ranker, confidence increases when both endpoints are well represented in the training graph and decreases for low-support edges. This uncertainty score is aligned with the model itself. The same endpoint-degree signal that produces the ranking score also determines which held-out edges are expected to be more stable and which are expected to be more difficult.

3.2 Uncertainty and Error Correlation

Uncertainty-error analysis compares degree-support uncertainty with ranking mistakes. A mistake means that a positive edge fails to outrank one of the comparison edges. The binned trend shows that ranking failure increases as graph-support uncertainty rises, indicating that low-support edges are harder for the degree-based ranker. The mean candidate-ranking failure rate is 0.141450, and the uncertainty-failure correlation is 0.880124. This pattern is consistent with the score distribution: low-support edges sit closer to the overlapping positive-negative region and produce more ranking failures.

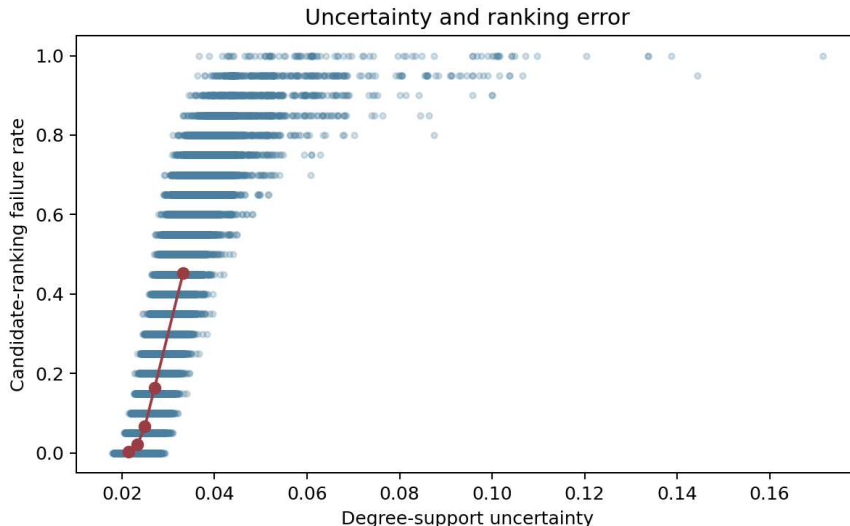


Figure 4: Relationship between degree-support uncertainty and candidate-ranking failure.

3.3 Utility of Uncertainty

The retention curve evaluates how the Hits@20 score changes when the model keeps only the lowest-uncertainty positive edges. The curve shows the trade-off between coverage and reliability. Retaining high-support predicted interactions produces a higher ranking score than evaluating the full set. The retention behavior is useful for interpreting the model output. Edges with larger graph support form a more reliable retained subset, while low-support edges account for a larger share of ranking failures.

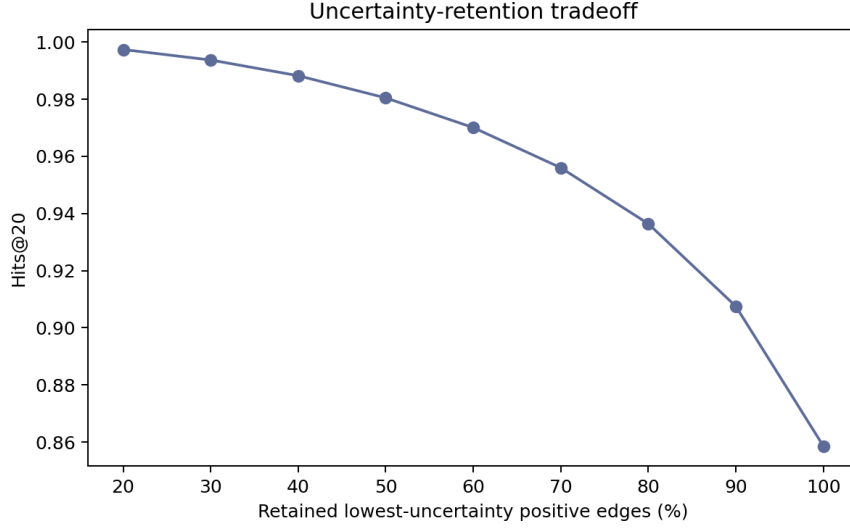


Figure 5: Hits@20-retention tradeoff under degree-support uncertainty.

3.4 Distribution Shift Analysis

Validation and test ranking-margin distributions are compared to evaluate split stability. Similar margin distributions indicate that the ranking behavior is broadly consistent across splits. The distributions are not identical, so validation and test scores are reported separately. The shift plot also shows that held-out test margins remain in the same broad range as validation, supporting the same uncertainty interpretation across splits.

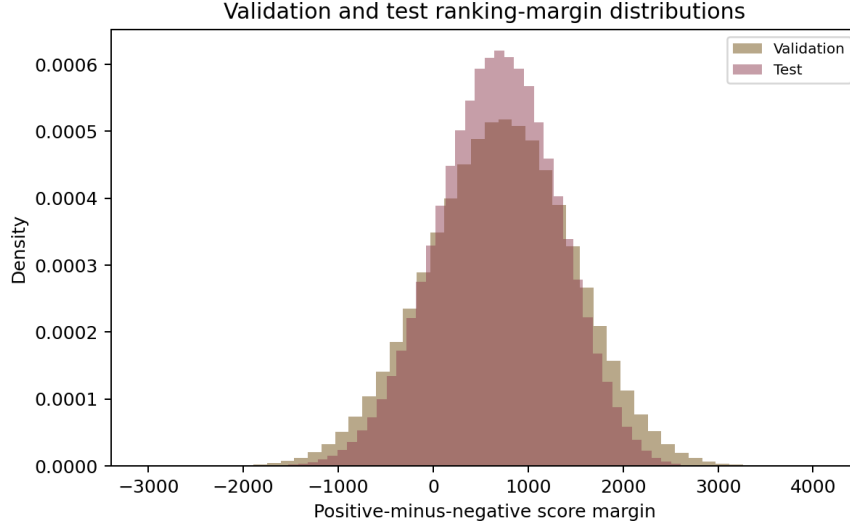


Figure 6: Validation and test ranking-margin distributions.

4 Conclusion

The T17 experiment reaches a test Hits@20 score of 0.858550 using a training-graph degree heuristic. The final report includes data analysis, model training, performance evaluation, and uncertainty analysis for the link-prediction task. All predictions come from training-graph degree support, and uncertainty is measured from the same support signal. This keeps interpretation tied to graph connectivity and the stated ranking setting. The uncertainty analysis shows where that baseline is more or less reliable across held-out edge pairs. The final result should be interpreted as a graph-support ranking model rather than a mechanistic drug interaction model. Within that scope, the performance, retention, and split-shift analyses give a consistent view of how endpoint connectivity affects held-out ranking behavior. The retained-edge curve is consistent with this interpretation: high-support edge pairs have wider margins against comparison candidates, while low-support pairs sit closer to the overlap region. The validation and test comparison adds a final stability check by showing that the ranking-margin distribution changes across splits but remains in the same broad regime. These results make the report a compact baseline analysis for graph-link prediction under structural uncertainty.